No.

Date.

Ex.4.7 | Ch.4 | P120 | Chloe | 2025.7.30

证明：(1) 推导 $Y_l^l(\theta,\phi)$

$P_l^m(x)=(-1)^m(1-x^2)^{m/2}\frac{d^m}{dx^m}P_l(x)$，当 $m=l$ 时，需计算 $\frac{d^l}{dx^l}P_l(x)$

$P_l(x)=\frac{1}{2^l l!}\frac{d^l}{dx^l}(x^2-1)^l=\frac{1}{2^l l!}\frac{(2l)!}{l!}x^l=\frac{(2l)!}{2^l(l!)^2}x^l$

$\frac{d^l}{dx^l}P_l(x)=\frac{(2l)!}{2^l(l!)^2}\cdot\frac{l!}{0!}x^0=\frac{(2l)!}{2^l l!}$

故 $P_l^l(x)=(-1)^l(1-x^2)^{l/2}\frac{d^l}{dx^l}P_l(x)=(-1)^l(1-x^2)^{l/2}\cdot\frac{(2l)!}{2^l l!}$

$P_l^l(\cos\theta)=(-1)^l\sin^l\theta\frac{(2l)!}{2^l l!}$

验证 当 $l=1$ 时 $P_1^1(\cos\theta)=-\sin\theta$，与表4.2一致

(2) 求 $Y_l^l(\theta,\phi)=\sqrt{\frac{(2l+1)}{4\pi}\cdot\frac{1}{(2l)!}}\,e^{il\phi}\cdot(-1)^l\sin^l\theta\frac{(2l)!}{2^l l!}$

$Y_3^2(\theta,\phi)=\sqrt{\frac{105}{32\pi}}\sin^2\theta\cos\theta e^{2i\phi}=N_3^2\,15\cos\theta\sin^2\theta e^{i2\phi}$

(3) 验证 $Y_3^2(\theta,\phi)$ 满足角方程 $\sin\theta\frac{\partial}{\partial\theta}(\sin\theta\frac{\partial Y}{\partial\theta})+\frac{\partial^2 Y}{\partial\phi^2}=-l(l+1)\sin^2\theta\cdot Y$

右边：代入 $l=3$，右边 $=-12\sin^2\theta Y$

左边：第一项 $\sin\theta\frac{\partial}{\partial\theta}(\sin\theta\frac{\partial Y}{\partial\theta})$，第二项 $\frac{\partial^2 Y}{\partial\phi^2}$

$Y=\cos\theta\sin^2\theta e^{i2\phi}$，$\frac{\partial Y}{\partial\theta}=e^{i2\phi}\sin\theta(2\cos^2\theta-\sin^2\theta)$

令 $H=\sin\theta\frac{\partial Y}{\partial\theta}=e^{i2\phi}\sin^2\theta(2\cos^2\theta-\sin^2\theta)$

$\frac{\partial H}{\partial\theta}=e^{i2\phi}4\sin\theta\cos\theta(\cos^2\theta-2\sin^2\theta)$

故左边 $=e^{i2\phi}\cdot4\sin^2\theta\cos\theta(\cos^2\theta-2\sin^2\theta)+(-4Y)$ $(Y=\cos\theta\sin^2\theta e^{i2\phi})$

$=4\cos\theta\sin^2\theta e^{i2\phi}[(\cos^2\theta-2\sin^2\theta)-1]$

$=-12\sin^4\theta\cos\theta e^{i2\phi}=-12\sin^2\theta Y=$ 右边

因此，当 $l=3$，$m=2$ 时，球谐函数 $Y_3^2(\theta,\phi)$ 满足角方程 (4.18)